

Air University Islamabad Department of Computer Science



LAB-14

SUBMITTED 19 May, 2026

Web Technologies Lab - Sir. Mohabbat Ali

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Title: ASP.NET CORE WEB APPLICATION DEVELOPMENT USING VISUAL STUDIO CODE

Objective:

By the end of this lab, students will be able to:

- Configure ASP.NET Core development environment in VS Code
- Create and run web applications using C# and ASP.NET Core
- Use Razor Pages for dynamic web content
- Handle form inputs using ASP.NET Core
- Build simple interactive web applications (calculator, login, forms, etc.)

Requirements:

- Visual Studio Code — Code Editor
- .NET SDK — For compiling and running applications
- Web Browser (Chrome / Edge) — For testing output
- Windows OS — Recommended environment

Install Extensions:

- C# (Microsoft)
- C# Dev Kit
- .NET Install Tool

System Setup Steps:

Step 1: Install Required Software

- Install Visual Studio Code
- Install .NET SDK
- Install required extensions in VS Code

Step 2: Verify .NET Installation

Open terminal:

```
dotnet --version
```

✓ If version appears → installation successful

Step 3: Create ASP.NET Core Project

```
dotnet new webapp -n MyFirstWebApp
cd MyFirstWebApp
code .
```

Step 4: Run Project

```
dotnet run
```

Open browser:

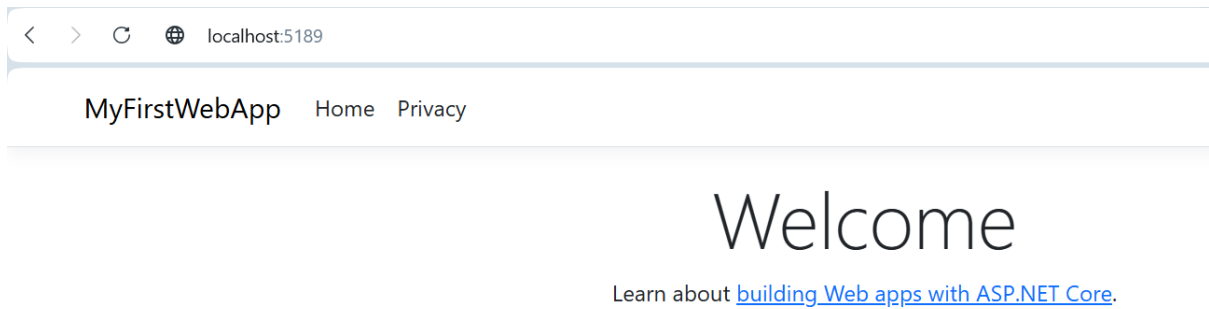
<https://localhost:5001>

```
● PS C:\Users\Mr.Khan\Desktop\LAB-14> dotnet new webapp -n MyFirstWebApp
The template "ASP.NET Core Web App (Razor Pages)" was created successfully.
This template contains technologies from parties other than Microsoft, see https://aka.ms/aspnetcore/10.0-third-party-notices for details.

Processing post-creation actions...
Restoring C:\Users\Mr.Khan\Desktop\LAB-14\MyFirstWebApp\MyFirstWebApp.csproj:
Restore succeeded.
```

```
○ PS C:\Users\Mr.Khan\Desktop\LAB-14> 
```

```
● PS C:\Users\Mr.Khan\Desktop\LAB-14> cd MyFirstWebApp
● PS C:\Users\Mr.Khan\Desktop\LAB-14\MyFirstWebApp> code .
○ PS C:\Users\Mr.Khan\Desktop\LAB-14\MyFirstWebApp> dotnet run
Using launch settings from C:\Users\Mr.Khan\Desktop\LAB-14\MyFirstWebApp\Properties\launchSettings.json...
Building...
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5288
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\Mr.Khan\Desktop\LAB-14\MyFirstWebApp
```



LAB ACTIVITIES

Activity 1: First Web Page

Objective

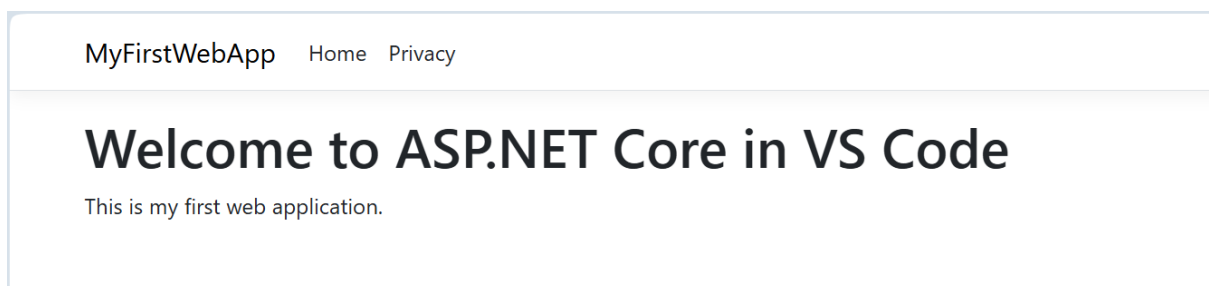
Run and customize default ASP.NET Core web app.

Steps:

Edit Pages/Index.cshtml:

```
<h1>Welcome to ASP.NET Core in VS Code</h1>
<p>This is my first web application.</p>
```

Output:



Activity 2: Simple Form Input

Objective

Create a form to take user input.

```
<form method="post">
  <input type="text" name="username" placeholder="Enter Name">
  <button type="submit">Submit</button>
</form>
```

```
@if (Request.Method == "POST")
{
  <p>Hello @Request.Form["username"]</p>
}
```

< > ↻ 🌐 localhost:5189

MyFirstWebApp Home Privacy

Hello Imad Khan

Activity 3: Simple Calculator

Objective

Perform addition using Razor Pages.

```
<form method="post">
  <input type="number" name="a">
  <input type="number" name="b">
  <button type="submit">Add</button>
</form>
```

```
@if (Request.Method == "POST")
{
    int a = int.Parse(Request.Form["a"]);
    int b = int.Parse(Request.Form["b"]);
    <p>Result: @(a + b)</p>
}
```

MyFirstWebApp Home Privacy

Simple Calculator

Result: 21

Activity 4: Login Validation

Objective

Create simple login logic.

```
<form method="post">
  <input type="text" name="user">
  <input type="password" name="pass">
  <button type="submit">Login</button>
</form>
```

```
@if (Request.Method == "POST")
{
  if (Request.Form["user"] == "admin" && Request.Form["pass"] == "123")
  {
    <p>Login Successful</p>
  }
  else
  {
    <p>Invalid Credentials</p>
  }
}
```

MyFirstWebApp [Home](#) [Privacy](#)

Username	Password	Login
Invalid Credentials		

LAB TASK

Task 1

Create a form that:

- Takes student name
- Takes course name
- Shows confirmation message

MyFirstWebApp

Home

Privacy

Student Registration Form

Enter Student Name

Enter Course Name

Submit

Student Imad Khan enrolled in Web Tech

Task 2

Build a page that:

- Takes two numbers
- Displays the largest number

MyFirstWebApp

Home

Privacy

Find Largest

Largest Number: 22

Task 3

Create a grade calculator:

- Input marks
- Output grade (A, B, C, Fail)

Grade A

Task 4

Design a feedback form:

- Name
- Message
- Display submitted feedback

Feedback Submitted

Name: Imad Khan

Message: Good! Very

Task 5

Create a digital clock page:

- Show current date and time using C#

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Digital Clock

Current Date and Time:

19/05/2026 9:35:26 pm